

Where Should You Work? A Data-Driven Location Choice Tool for Early-Career Workers

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1. Topic or Question

This project examines how differences in wages, the cost of living, and job market conditions across U.S. states affect the real outcomes of early-career workers. A higher salary does not always mean a better standard of living, since prices and housing costs vary widely by location. Many recent graduates are flexible in where they work, but they often focus on salary or job opportunities without fully considering differences in affordability. This project aims to build a data-driven tool that helps users compare locations and understand where their income will actually go further after accounting for living costs.

2. Intended audience or users

The main users of this app are early-career workers and recent graduates who are deciding where to start their careers. This is especially true for international students, who often have fewer geographic constraints but may be less familiar with differences in living conditions across states. Because these users are typically flexible in where they work, they are more likely to benefit from comparing locations in a structured way rather than focusing only on salary or job availability. The app is intended to help them make more informed decisions by showing how income, rent, and local conditions differ across places.

3. Likely data source

This project will combine several public datasets to compare income, cost of living, and job market conditions across U.S. states.

Wage data will come from the Bureau of Labor Statistics, using the Occupational Employment and Wage Statistics (OEWS). This allows the app to show how salaries differ across locations, and it can also reflect differences across occupations.

To adjust for differences in prices, the project will use regional price parity data from the Bureau of Economic Analysis. This makes it possible to convert nominal wages into a measure of real purchasing power, which is more meaningful when comparing different states.

Housing costs will be included using median gross rent data from the U.S. Census Bureau. Rent is one of the largest expenses for recent graduates, so including it helps make the comparison more realistic.

Finally, unemployment rates from the Bureau of Labor Statistics will be used to reflect local labor market conditions. This adds another dimension to the comparison, since some areas may have higher wages but weaker job markets.

Together, these data sources allow the app to compare not just salaries, but also affordability and job opportunities across locations.

4. Proposed application or dashboard

Users will be able to select one or more states and either input an expected salary or choose from a set of income levels. Based on these inputs, the app will compare how income translates into actual living conditions across locations. In particular, it will show how much of a user's income would likely be spent on rent, and how much remains after major expenses.

The app will present this information in a few simple views. One part will allow users to compare different states side by side, including salary levels, rent, and adjusted measures of affordability. Another part will rank locations based on how favorable they are given the user's income, helping users quickly identify which states offer better overall outcomes. The app will also include basic labor market information, such as unemployment rates, to give some sense of how strong job opportunities are in each location.

5. Main indicators and visualization

The main indicators in the app include wages, median rent, and unemployment rates, along with simple measures that combine these variables to reflect affordability. Bar charts will be used to compare income and rent across states, and a ranking display will highlight which locations perform better based on the user's selected income level. The app will also include a simple comparison between income and rent to show how much of a user's earnings would likely go toward housing. Depending on data availability, the app could also be extended to include additional factors such as state income taxes, food varieties, and weather.

6. *Potential Data*

1. Different States' Median Gross Rent by Bedrooms
[https://data.census.gov/table/ACSDT1Y2024.B25031?q=median+gross+rent&t=Housing&g=010XX00US\\$0400000_040XX00US41,44,45,46,47,48&y=2024](https://data.census.gov/table/ACSDT1Y2024.B25031?q=median+gross+rent&t=Housing&g=010XX00US$0400000_040XX00US41,44,45,46,47,48&y=2024)
2. Occupational Employment and Wage Statistics
<https://www.bls.gov/oes/tables.htm>
3. Unemployment Rates for States
<https://www.bls.gov/web/laus/laumstrk.htm>
4. Real Personal Income for States
<https://bea.gov/data/income-saving/real-personal-income-states>
5. Regional Price Parities by State and Metro Area
<https://www.bea.gov/data/prices-inflation/regional-price-parities-state-and-metro-area>